

THE UNIVERSITY OF EDINBURGH

School of Mathematics



Laptop Manufacturing System

Simulation

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1 Introduction

This report presents a discrete-event simulation study of a laptop manufacturing system, developed using Simul8. The model captures the full production process from order arrival through motherboard and case manufacturing, workbench assembly, quality assurance, and final boxing, alongside inventory management for aluminium blocks and hard disks. For processes with available empirical data, input distributions were fitted using maximum likelihood estimation and validated using standard statistical goodness-of-fit procedures. Steady-state performance was estimated via the batch means method following a warm-up period of 1,240 minutes identified through Welch’s graphical procedure. The study aims to identify the system’s key bottlenecks and evaluate targeted improvement scenarios to reduce average cycle time and increase order fulfilment.

2 Input Analysis

The laptop manufacturing system is driven by multiple sources of variability, including stochastic order arrivals, processing times across several assembly stages, and probabilistic rework and inventory mechanisms. Since system performance is highly sensitive to both the magnitude and structure of this variability, careful input modelling is essential to ensure credible simulation results.

The input model is constructed by combining two sources of information. Several processes are specified directly in the project brief, including Uniform, Triangular, Beta-based, and deterministic processing times, as well as discrete order-size and rework mechanisms. In contrast, four key processes; interarrival times, initial phase service times, keyboard and mouse placement times, and case assembly times, are supported by empirical data and require statistical modelling.

For these processes, a structured modelling approach is adopted. Candidate distribution families are first selected based on data characteristics and process understanding, with particular attention to support and skewness. Parameters are then estimated using maximum likelihood, and competing models are assessed using a combination of graphical diagnostics (histograms and Q–Q plots) and statistical measures, including information criteria and goodness-of-fit tests. Final model selection is based on overall consistency with the data, physical realism, and suitability for simulation, rather than reliance on any single test.

Data Verification and Independence

Prior to fitting, each dataset was checked for completeness and physical plausibility. No missing values or non-positive observations were found in any of the four data sets, indicating that the data is suitable for modelling without the need for cleaning or adjustment. Sample autocorrelation functions (ACFs) were then computed for all four variables up to lag 30 to assess the independently and identically distributed (i.i.d.) assumption required for the fitting procedure. The 95% confidence bounds for the autocorrelations under the null hypothesis of independence are $\pm 1.96/\sqrt{200} \approx \pm 0.139$. All coefficients at every lag fell within this band for every variable (Figure 1), providing no evidence of material serial dependence. The i.i.d. assumption is therefore retained for all four data sets.

Exploratory Data Analysis

Histograms with kernel density estimates and boxplots are shown in Figures 2 and 3, with summary statistics reported in Table 1.

Analysis of the historical data reveals distinct patterns for each manufacturing stage. The interarrival times exhibit clear right-skewness of 1.54 and a coefficient of variation $CV = 0.94$ which is close to the unit value associated with the Exponential distribution. The Initial Phase of production is highly consistent and predictable. It shows very little variation between items with times being approximately symmetric with a skewness of -0.025 and low variability of $CV = 0.105$. This indicates a tightly concentrated unimodal distribution. The Keyboard and Mouse times are strongly right-skewed with a skewness of 2.30, kurtosis of ≈ 11.79 with a pronounced long tail extending to 38.8 minutes, suggesting a heavy-tailed distribution such as the Lognormal. The Case Assembly times are also approximately symmetric with a skewness of -0.028 and moderate variability of $CV = 0.317$.

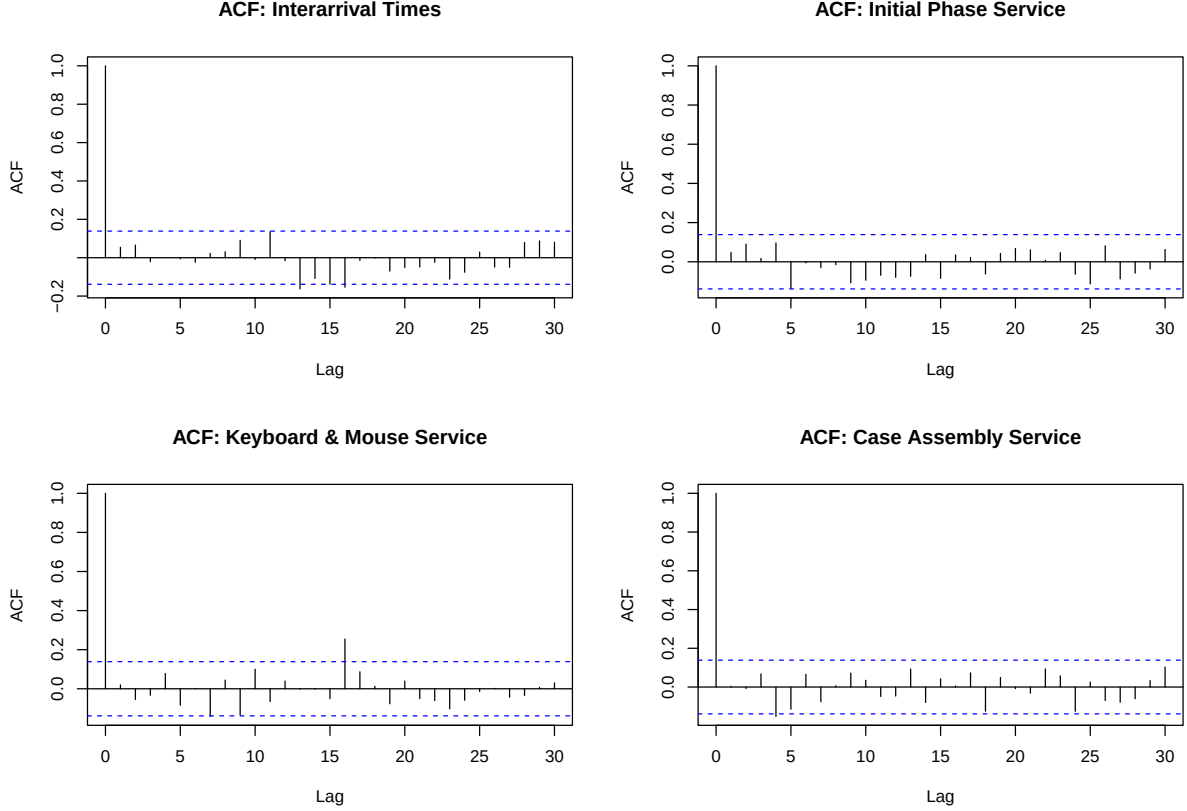


Figure 1: Sample autocorrelation functions for each empirical input variable up to lag 30. All coefficients fall within the 95% confidence band (± 0.139).

Table 1: Descriptive statistics for empirical input variables ($n = 200$).

Variable	Mean	S^2	S	Min	Median	Max	Skewness	CV
Interarrival Times	8.145	58.70	7.661	0.082	5.431	42.242	1.542	0.941
Initial Phase Service	5.236	0.302	0.550	3.718	5.220	6.791	-0.025	0.105
Keyboard & Mouse Service	7.729	25.379	5.038	1.274	6.362	38.768	2.301	0.652
Case Assembly Service	3.272	1.075	1.037	0.559	3.290	5.668	-0.028	0.317

Exploratory Data Analysis: Histograms with Kernel Density

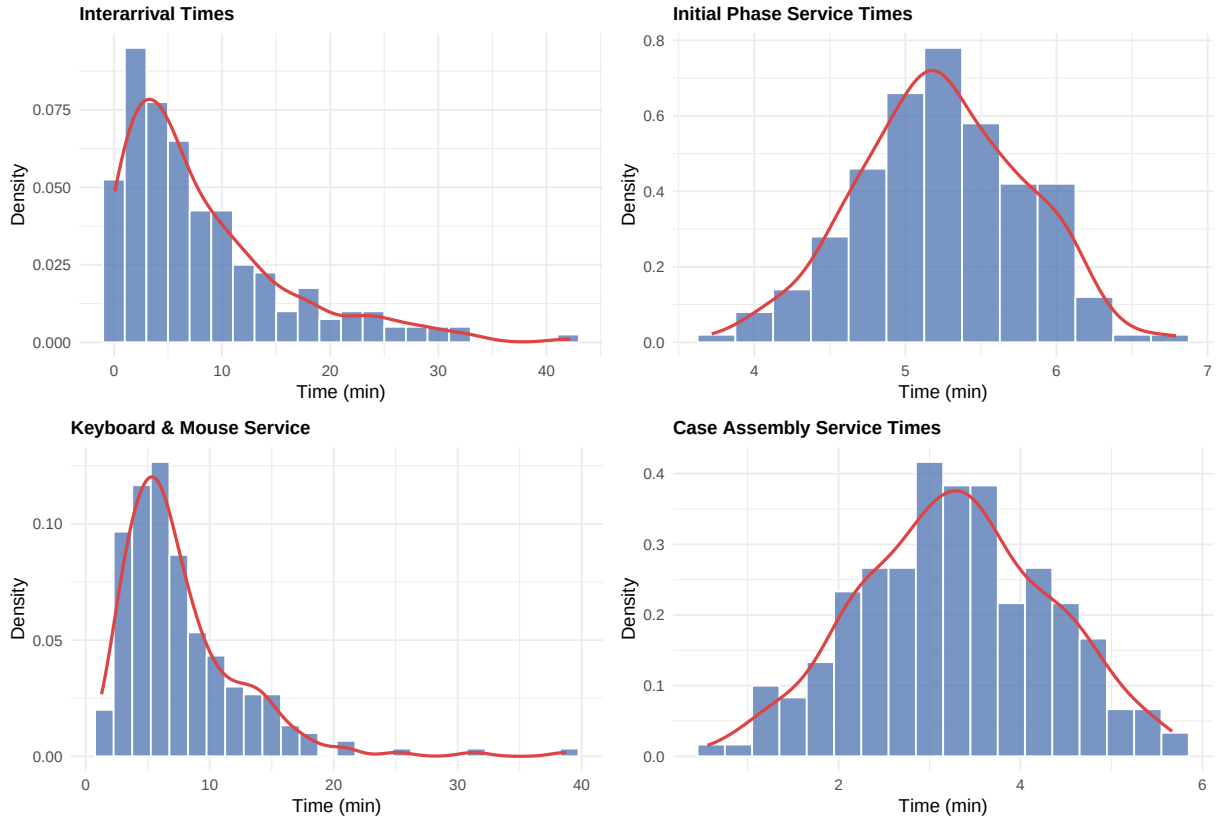


Figure 2: Histograms with kernel density estimates for the four empirical input variables.

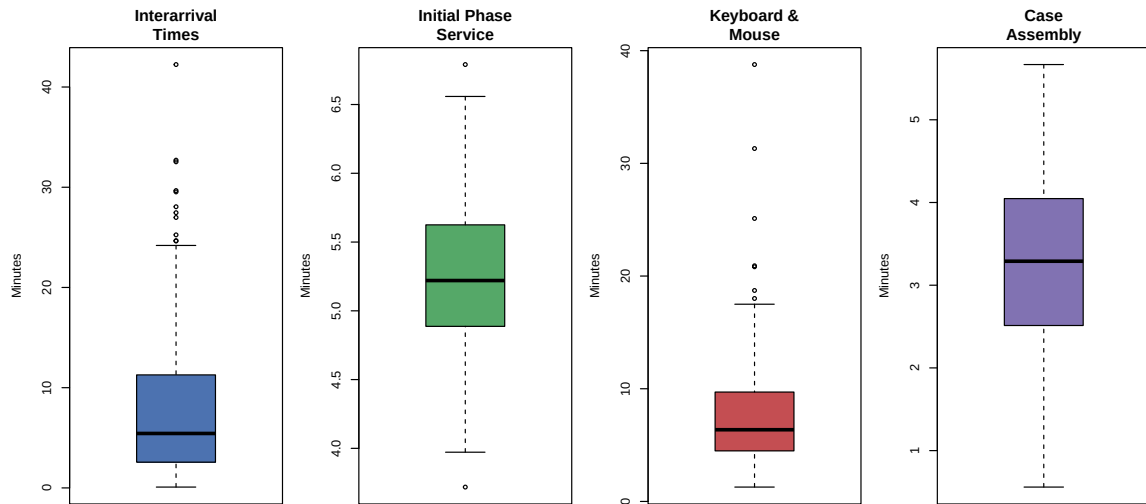


Figure 3: Boxplots illustrating the central tendency, spread, and tail behaviour of each empirical variable.

Distribution Fitting and Parameter Estimation

For each variable, candidate distribution families were selected based on both process considerations and the observed data characteristics, particularly skewness and support. Parameters were then estimated using maximum likelihood. Competing models were evaluated using information criteria (AIC and BIC), supported by visual diagnostics including density overlays and Q-Q plots.

The Akaike Information Criterion, $AIC = 2k - 2 \ln \hat{L}$, was used to compare non-nested models, with lower values indicating a better balance between goodness-of-fit and model complexity. Differences of fewer than two AIC units were treated as inconclusive, in which case model selection was guided by consistency with the physical properties of the process, particularly that all processing times must be non-negative.

Interarrival Times.

The project brief indicates that order interarrival times follow an exponential distribution, consistent with the relationship between Poisson arrivals and exponential inter-event times. This assumption is supported by the data: the observed coefficient of variation ($CV = 0.94$) is close to the unit value characteristic of the Exponential distribution, and the histogram exhibits the expected right-skewed shape.

The rate parameter was estimated by maximum likelihood as $\hat{\lambda} = 1/\bar{X}$, giving

$$\hat{\lambda} = \frac{1}{8.1445} = 0.1228 \text{ min}^{-1}(\text{per minute}).$$

The Exponential distribution achieves the lowest AIC (1240.938) and BIC (1244.236) among the candidate models, and shows good agreement with the data in both the density overlay (Figure 4) and Q-Q plot (Figure 5). These results support its use as the input model for interarrival times.

Initial Phase Service Times.

The near-zero skewness (-0.025) and low coefficient of variation ($CV = 0.105$) indicate a symmetric and tightly concentrated distribution making the Normal distribution a natural initial candidate. However, positive-support alternatives, including the Gamma distribution, were also considered to ensure consistency with the non-negative nature of service times.

The Normal distribution achieves the lowest AIC (331.213), with the Gamma close behind (332.849), giving a difference of 1.64 units and therefore no strong statistical preference between the two models. The K-S statistic for the Gamma ($D = 0.0256$, $p = 0.9994$) is also slightly better than that of the Normal ($D = 0.0379$, $p = 0.9360$), although both provide an excellent fit.

Given the comparable statistical performance, the Gamma distribution is selected as it ensures strictly non-negative values while maintaining a very close fit to the data. The estimated parameters imply a mean and variance consistent with the sample statistics, and the large shape parameter results in an approximately bell-shaped distribution, closely matching the observed data. Fitting the initial phase data resulted in a shape parameter $\beta = 89.76$ and a scale parameter $\theta = 17.14$.

Keyboard and Mouse Placement Service Times.

This variable exhibits strong right-skewness ($\text{skewness} = 2.30$) and a pronounced long tail, with values extending to 38.8 minutes, indicating substantial variability in task duration. Such behaviour is consistent with a multiplicative process in which occasional complex assemblies result in significantly longer service times, making the Lognormal distribution a natural candidate.

The Lognormal distribution provides a substantially better fit than all alternatives, achieving the lowest AIC (1103.646) by a margin of 13.8 units over the next best model (Gamma, $AIC = 1117.419$), constituting strong evidence in its favour. Visual diagnostics (Figure 8 and Figure 9) confirm excellent agreement across both the body and upper tail of the data.

The fitted parameters are $\hat{\mu}_{\ln} = 1.8742$ and $\hat{\sigma}_{\ln} = 0.5804$, implying a mean of approximately 7.73 minutes on the original scale, consistent with the sample mean. The Lognormal distribution is therefore selected as the input model for this process.

Case Assembly Service Times.

The near-zero skewness (-0.028) and moderate coefficient of variation ($CV = 0.317$) indicate an approximately symmetric distribution with moderate variability. The Normal distribution is therefore a natural candidate. However, positive-support alternatives, including the Weibull and Gamma distributions, were also considered to ensure consistency with the non-negative nature of processing times.

The Weibull distribution achieves the lowest AIC (583.334), with the Normal close behind (584.992), giving a difference of 1.66 units and therefore no strong statistical preference between the two models. Visual diagnostics (Figure 10 and Figure 11) show that both distributions provide a good fit to the data.

Given the comparable statistical performance, the Weibull distribution is selected as it ensures strictly non-negative values while maintaining an excellent fit. The fitted parameters are $\hat{\beta} = 3.526$ (shape) and $\hat{\alpha} = 3.636$ (scale), implying a mean of approximately 3.27 minutes, consistent with the sample mean.

Table 2: AIC and BIC comparison across candidate distributions. Bold denotes the selected model. Asterisks (*) denote distributions excluded on physical grounds (support includes negative values).

Variable	Distribution	AIC	BIC
Interarrival Times	Exponential	1240.938	1244.236
	Weibull	1242.276	1248.872
	Gamma	1242.394	1248.991
	Lognormal	1275.131	1281.728
Initial Phase Service	Gamma	332.849	339.446
	Normal*	331.213	337.809
	Lognormal	334.734	341.330
	Weibull	343.565	350.161
Keyboard & Mouse Service	Lognormal	1103.646	1110.243
	Gamma	1117.419	1124.016
	Weibull	1141.073	1147.669
	Exponential	1219.982	1223.280
Case Assembly Service	Weibull	583.334	589.931
	Normal*	584.992	591.588
	Gamma	601.054	607.650
	Lognormal	623.106	629.703

*Normal has support $(-\infty, +\infty)$ and is excluded for service time variables.

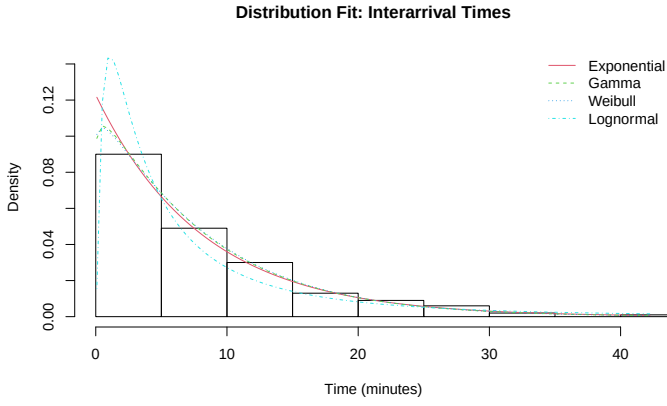


Figure 4: Density overlay: Interarrival Times. The Exponential curve closely tracks the empirical distribution across the full range.

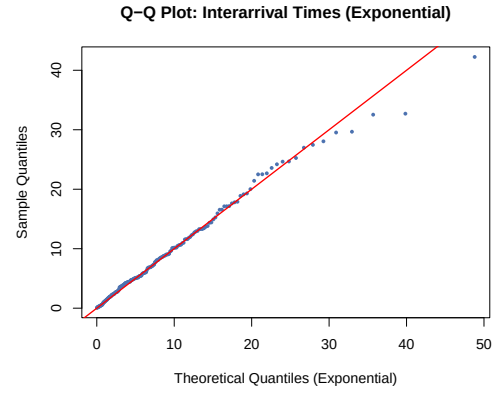


Figure 5: Q-Q plot: Interarrival Times against the fitted Exponential distribution.

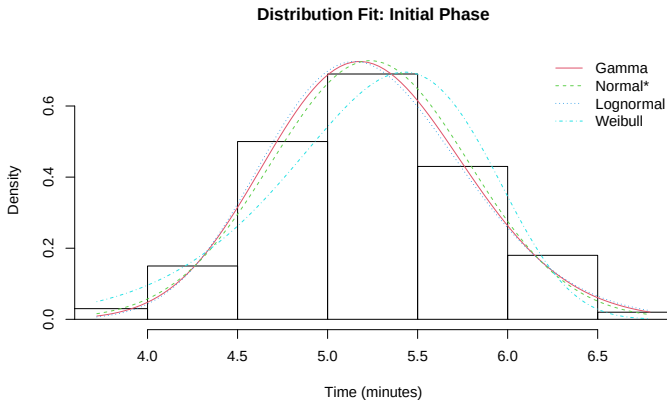


Figure 6: Density overlay: Initial Phase Service Times. The Gamma curve provides the best physically valid fit; Normal is shown for reference only.

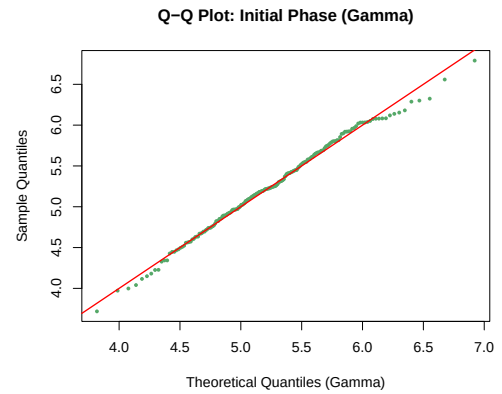


Figure 7: Q-Q plot: Initial Phase Service Times against the fitted Gamma distribution.

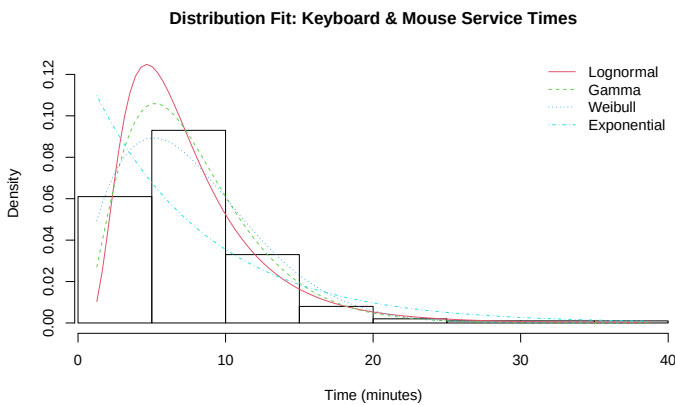


Figure 8: Density overlay: Keyboard and Mouse Service Times. The Lognormal curve captures the heavy right tail decisively better than all alternatives.

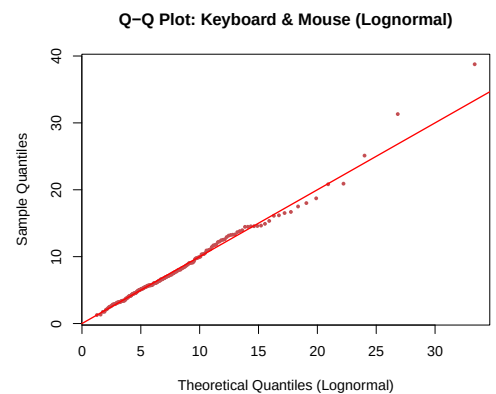


Figure 9: Q-Q plot: Keyboard and Mouse Service Times against the fitted Lognormal distribution.

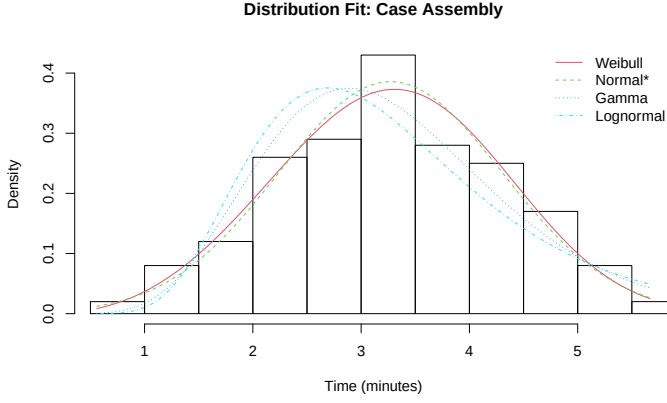


Figure 10: Density overlay – Case Assembly Service Times. The Weibull curve achieves the lowest AIC and is physically bounded; Normal is shown for reference only.

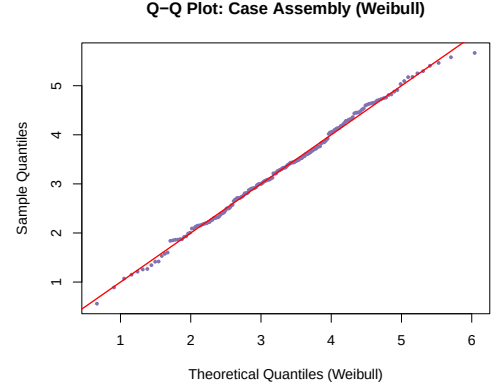


Figure 11: Q-Q plot: Case Assembly Service Times against the fitted Weibull distribution.

Goodness-of-Fit Validation

The Kolmogorov–Smirnov (K–S) test was used as the formal goodness-of-fit measure, comparing the empirical distribution with the fitted theoretical models. For all four variables, the computed K–S statistics (D_n) are well below the corresponding critical value for $n = 200$ at the 5% significance level, and the associated p -values are comfortably above 0.05 as evidenced in Table 3.

It is noted that, since parameters are estimated from the data, the K–S test is conservative in this setting; consequently, the results should be interpreted as supportive rather than definitive. On this basis, there is no statistical evidence to reject any of the selected distributions.

These findings are reinforced by the Q–Q plots (Figures 5 to 11), which show close agreement between empirical and theoretical quantiles across the full range of the data, with no systematic deviation in either the body or the tails.

Table 3: Kolmogorov-Smirnov goodness-of-fit results for selected distributions ($n = 200$, $\alpha = 0.05$, $D_{0.05} = 0.0962$).

Variable	Distribution	D_n	p -value	Decision
Interarrival Times	Exponential	0.0474	0.7589	Do not reject H_0
Initial Phase Service	Gamma	0.0256	0.9994	Do not reject H_0
Keyboard & Mouse Service	Lognormal	0.0293	0.9955	Do not reject H_0
Case Assembly Service	Weibull	0.0274	0.9982	Do not reject H_0

Final Input Model Specification

Table 4 consolidates the complete set of input distributions for the simulation model, combining the four empirically fitted distributions with those specified directly in the project description.

Table 4: Complete input model specification for the laptop manufacturing simulation.

Process	Distribution	Parameters	Source
Empirically fitted ($n = 200$ observations)			
Interarrival Times	Exponential	$\lambda = 0.1228 \text{ min}^{-1}$	Fitted
Initial Phase	Gamma	$\beta = 89.76, \theta = 17.14$	Fitted
Keyboard & Mouse	Lognormal	$\mu_{\ln} = 1.874, \sigma_{\ln} = 0.582$	Fitted
Case Assembly	Weibull	$\beta = 3.526, \alpha = 3.636 \text{ min}$	Fitted
Specified in the project description			
Power Assembly	Uniform	$U(2, 6) \text{ min}$	Specified
Main Part Assembly	Triangular	$\text{Tri}(5, 8, 15) \text{ min}$	Specified
Shaping the Screen	Uniform	$U(3, 5) \text{ min}$	Specified
Placing the Screen	Uniform	$U(4, 10) \text{ min}$	Specified
Cutting – Keyboard	Uniform	$U(5, 8) \text{ min}$	Specified
Cutting Alum. Plates	Deterministic	8 min	Specified
Hard Disk Assembly	Transformed	$3X + 1 \text{ min}, X \sim \text{Beta}(3, 2)$	Specified
Place Motherboard	Uniform	$U(4, 8) \text{ min}$	Specified
Quality Assurance	Triangular	$\text{Tri}(8, 14, 16) \text{ min}$	Specified
Boxing	Uniform	$U(4, 8) \text{ min}$	Specified
Order Batch Size	Shifted Bin.	$1 + \text{Bin}(5, 0.6)$	Specified

Input Analysis Assumptions

The input model is based on the following assumptions.

Independence: Observations within each variable are treated as independently and identically distributed (i.i.d.). This is supported by the ACF analysis up to lag 30, which shows no evidence of material serial dependence in any of the four series.

Stationarity: Arrival and service processes are assumed to be time-homogeneous throughout the simulation horizon. This is consistent with the project specification, which states continuous 24-hour operation with no shift variation.

Point parameter estimates: All distribution parameters are treated as fixed at their maximum likelihood estimates. The uncertainty associated with finite sample sizes is not explicitly modelled.

Specified distributions: Distributions defined directly in the project specification (e.g. Uniform, Triangular, Beta-based, and deterministic times) are adopted without further empirical validation.

3 Model Formulation

3.1 Overview

The simulation model replicates a simplified laptop manufacturing process built in Simul8. It tracks orders from arrival through motherboard and case manufacturing, workbench assembly, quality assurance, and final boxing. Inventory control for aluminium blocks and hard disks is also included. The complete model is shown in Figure 12.

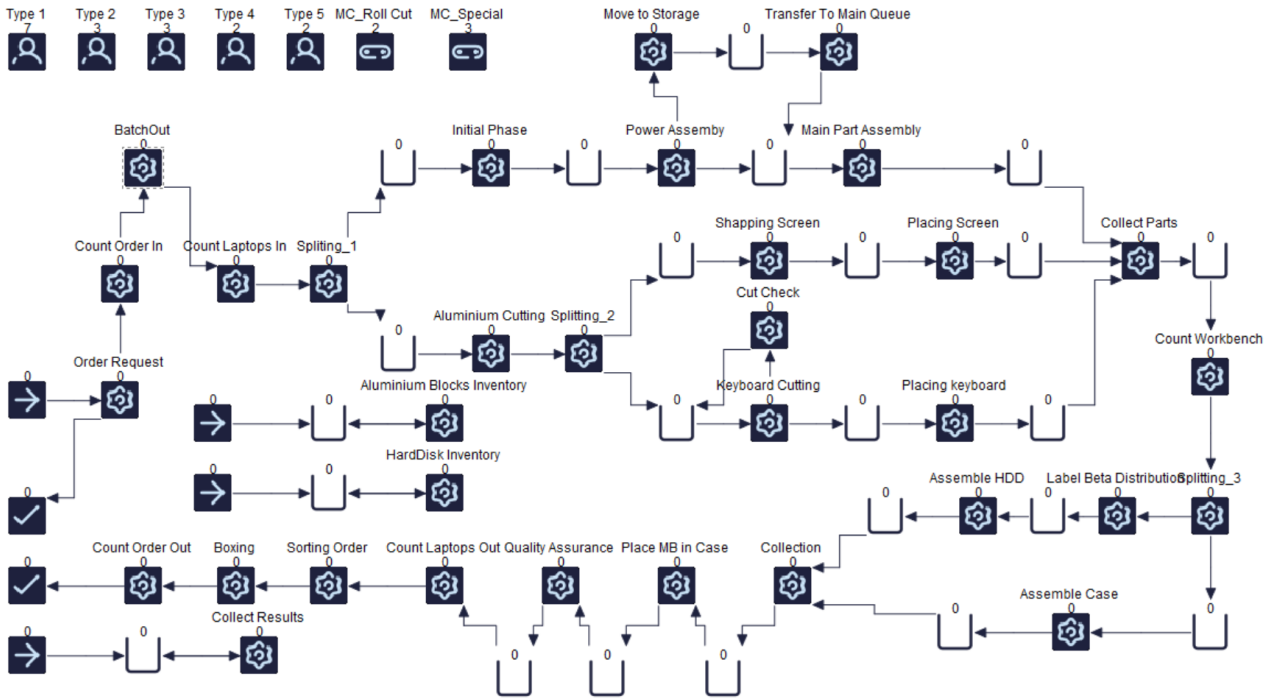


Figure 12: Complete Simul8 model of the laptop manufacturing process.

Each work item carries a set of labels throughout the simulation. These labels act like tags on each laptop, holding information such as when it arrived, which order it belongs to, and how many times it has been through rework. The key labels and global tracking variables are summarized in Tables 5 and 6.

Table 5: Labels carried by each work item.

Label	Purpose
lbl_ArrivalTime	Time the laptop entered the system
lbl_BatchSize	Number of laptops in the order
lbl_BetaHDD	Pre-sampled value for HDD assembly time
lbl_CountCut	Number of keyboard cutting attempts
lbl_CycleTime	Total time from entry to exit
lbl_DepartureTime	Time the laptop left the system
lbl_OrderID	Links all laptops in the same order
lbl_OrderSize	Total laptops requested in the order
lbl_RouteChoice	Controls routing at decision points

Table 6: Key global variables used across the model.

Variable	Reset	Purpose
<code>glb_AluminiumBlocks_Inventory</code>	50	Aluminium stock level
<code>glb_HDD_Inventory</code>	70	Hard disk stock level
<code>glb_TotalLaptopsInSystem</code>	0	Laptops currently in production
<code>glb_Order_Accepted</code>	0	Orders accepted
<code>glb_OrderCompleted</code>	0	Orders fully shipped
<code>glb_TotalCycleTime</code>	0	Cumulative cycle time
<code>glb_Average_CycleTime_10</code>	0	Rolling average of last 10 cycle times
<code>glb_TCT</code>	0	Cycle time accumulator per replication
<code>glb_Numb_Replication</code>	0	Replication counter
<code>glb_WorkbenchNumber</code>	2	Available workbenches

Three internal spreadsheets support the model: `glb_Data10` stores the last 10 cycle times for warm-up analysis, `glb_OrderID.Boxing` tracks orders waiting to be boxed, and `glb_Results1` records per-replication output for the confidence interval calculation.

3.2 Arrival Process and Order Management

Orders arrive with exponential inter-arrival times. From the 200 observations provided, the mean inter-arrival time was estimated as $\bar{x} = 8.1445$ minutes, giving rate $\hat{\lambda} = 0.123$ orders per minute.

Each order requests:

$$N = X + 1 \text{ laptops, } X \sim \text{Binomial}(n = 5, p = 0.6) \quad (1)$$

giving an average order size of 4 laptops. The company accepts an order only if the total laptops in production would not exceed 100:

$$\text{Accept} \iff \text{glb_TotalLaptopsInSystem} + \text{lbl_BatchSize} \leq 100 \quad (2)$$

Rejected orders exit through `End 1` immediately. Accepted orders pass through `Count Order In` (which timestamps `lbl_ArrivalTime`), then `Splitting_1` which creates one pair of work items per laptop, one for the motherboard line and one for the case line, each carrying the same `lbl_OrderID`.

3.3 Motherboard Manufacturing Line

Each laptop generates one work item that flows through three sequential motherboard stages (Table 7).

Table 7: Motherboard manufacturing stages.

Stage	Distribution	Parameters	Worker
Initial Phase	Gamma	$\alpha = 89.7784, \beta = 0.0583$ min	Type 1
Power Assembly	Uniform	$[2, 6]$ min	Type 1
Main Part Assembly	Triangular	min= 5, mode= 8, max= 15 min	Type 2

The queue before Power Assembly holds at most 3 items. When full, Initial Phase is blocked from releasing further work using `Block Current Routing` in the *Before Selecting* logic. The

queue before Main Part Assembly holds at most 2 items; when full, items are routed to a storage area via **Move to Storage** (2 min delay) and returned via **Transfer To Main Queue** (3 min delay), controlled by the **On Work complete** logic of Power Assembly which checks the downstream queue count before routing, as shown in Figures 13 and 14.

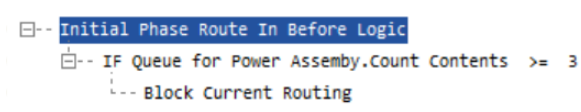


Figure 13: Blocking logic when Power Assembly queue is full.

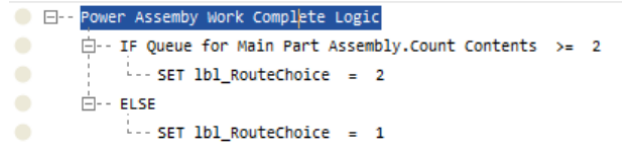


Figure 14: Routing logic at Power Assembly.

When items leave storage, the **Transfer To Main Queue** exit logic checks whether the Main Part Assembly queue has fewer than 2 items before moving them forward, if the queue is still full, the item is blocked in storage until space becomes available.

3.4 Case Manufacturing Line

In parallel, a second work item per laptop flows through the case manufacturing stages (Table 8).

Table 8: Case manufacturing stages.

Stage	Distribution	Parameters	Resource
Aluminium Cutting	Fixed	8 min	MC_Roll Cut
Shaping Screen	Uniform	[3, 5] min	Type 3, MC_Special
Placing Screen	Uniform	[4, 10] min	Type 1 Worker
Keyboard Cutting	Uniform	[5, 8] min	Type 3, MC_Special
Placing Keyboard	Lognormal	$\mu_{ln} = 1.874, \sigma_{ln} = 0.580$	Type 1 Worker

After aluminium cutting, **Splitting_2** sends one copy to the screen path (Shaping Screen → Placing Screen) and one copy to the keyboard path (Cut Check → Keyboard Cutting → Placing Keyboard).

Keyboard rework: The keyboard cutting step is imperfect. After each attempt the label **lbl_CountCut** is incremented and a probabilistic check determines whether the item must be recut. The rework probabilities are 0.2 on the first attempt, 0.1 on the second, and 0.0 on the third, after which all items pass unconditionally ($P(\text{rework}) = 0.2, 0.1, 0.0$ for **lbl_CountCut** = 1, 2, 3 respectively).

This logic sits in the **Cut Check** dummy activity using **RANDOM(1)** to generate the probability and **lbl_RouteChoice** to direct items back to cutting or forward. The Visual Logic is shown in Figure 17. The total number of rework events is recorded in **glb_CutsRepaired**.

Aluminium inventory: Each aluminium cutting job consumes one block. The stock level is tracked in **glb_AluminiumBlocks_Inventory** (starting at 50). When stock falls below $s = 10$, a replenishment order of $Q = 50$ blocks is triggered and arrives 900 minutes (15 hours) later. If stock reaches zero, **Aluminium Cutting** is blocked until replenishment arrives (Figure 15).

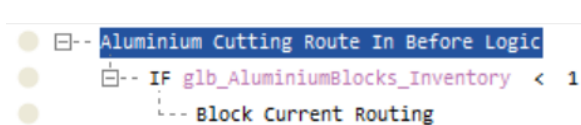


Figure 15: Blocking logic for Aluminium.

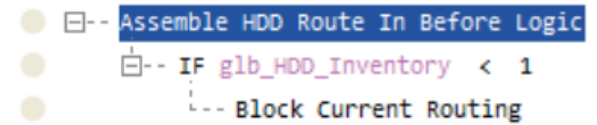


Figure 16: Routing logic at HDD Assembly.

3.5 Workbench Assembly

Assembly requires three components to arrive together, the motherboard, the upper body (with screen), and the lower body (with keyboard). With only 2 workbenches available for this process (`glb_WorkbenchNumber = 2`), the system must coordinate carefully.

The activity **Collect Parts** acts as the coordination gate using Simul8's *Collect* routing discipline with *Match on: lbl_OrderID* and *Assemble* enabled. This ensures Simul8 automatically waits until one item from each of the three feeding queues is available and that they belong to the same order before releasing them as a single assembled unit.

Once parts are collected, **Count Workbench** confirms a workbench is free, then **Splitting_3** routes the item into two simultaneous workbench tasks:

1. **HDD and Peripherals:** (Type 2 worker) - service time $T_{\text{HDD}} = 3X + 1$ minutes where $X \sim \text{Beta}(3, 2)$. The Beta sample is pre-computed at **Label Beta Distribution** and stored in `lbl_BetaHDD` before the item enters **Assemble HDD**.
2. **Case and Motherboard placement:** (Type 1 worker) - two sequential tasks without releasing the workbench:
 - **Assemble Case:** Weibull(3.526, 3.636) minutes
 - **Place MB in Case:** Uniform(4, 8) minutes

Both tasks converge at **Collection** (Collect routing) before the finished laptop moves to quality assurance.

Hard disk inventory: Each laptop consumes one hard disk at **Assemble HDD**. Stock is tracked in `glb_HDD_Inventory` (starting at 70). An $(s = 10, Q = 70)$ policy applies with 900 minute lead time. If stock reaches zero, **Assemble HDD** is blocked (Figure 16).

3.6 Quality Assurance, Boxing and Results Collection

Quality Assurance: is performed by a Type 4 worker with service time:

$$T_{\text{QA}} \sim \text{Triangular}(\text{min} = 8, \text{mode} = 14, \text{max} = 16) \text{ minutes} \quad (3)$$

Cycle time recording: As each laptop exits boxing at **Count Order Out**, the logic in Figure 18 is implemented:

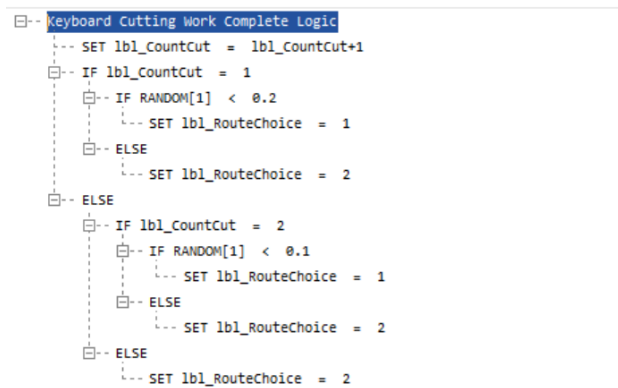


Figure 17: Rework decision logic at Cut Check.

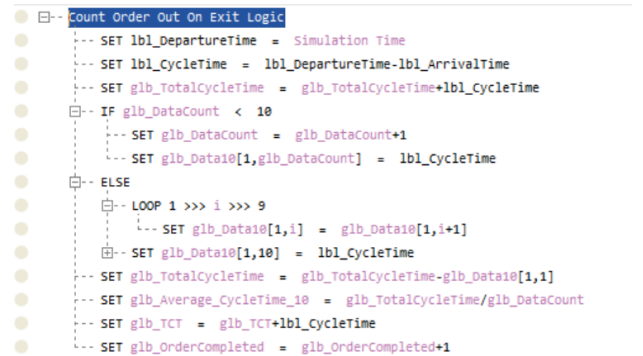


Figure 18: Exit logic at Count Order Out.

The rolling average `glb_Average_CycleTime_10` is used to determine the warm-up period graphically. The variable `glb_TCT` accumulates cycle time within each replication and is reset between replications.

Boxing: Completed laptops wait at `Sorting Order` until all members of their order have passed quality assurance. Once the full batch is present, `Boxing` processes the order in `Uniform(4,8)` minutes using one Type 5 worker. Matched items are deleted from the queue, leaving one representative item to exit. `Count Order Out` then decrements `glb.TotalLaptopsInSystem` to free capacity for incoming orders.

Results collection: The `Collect Results` activity fires at the end of each replication using a dedicated timed start point. As shown in Figure 19, it records the per-replication average cycle time and order count into `glb.Results1` and then resets the accumulators ready for the next run:

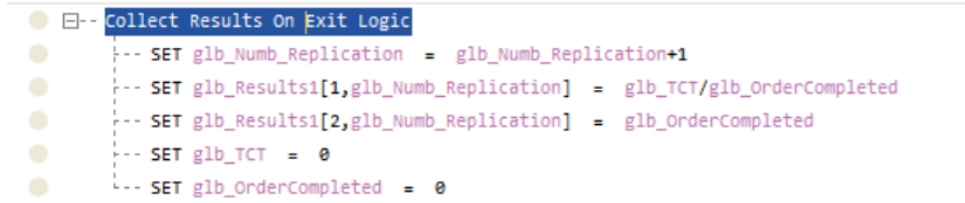


Figure 19: Visual Logic at `Collect Results` storing the replication average cycle time (`glb_TCT / glb_OrderCompleted`) and order count into `glb.Results1` before resetting for the next replication.

3.7 Model Assumptions

The following assumptions were made during model construction:

1. The factory is assumed to operate 24 hours continuously with no planned downtime.
2. Components at `Collect Parts` are matched by `lbl_OrderID` using Simul8's built-in Collect discipline, ensuring only parts from the same order are assembled together.
3. The Beta distribution for HDD assembly time is pre-sampled at `Label Beta Distribution` to ensure the formula $3X + 1$ is correctly applied.
4. Boxing begins only when all laptops in an order have passed quality assurance, orders ship as complete batches.
5. Only aluminium blocks and hard disks are tracked as constrained raw materials; all other components are assumed always available.

4 Output Analysis and Baseline Model Results

The purpose of this section is to estimate the steady-state average cycle time of a laptop in the manufacturing system using appropriate statistical procedures, and to ensure that the resulting estimate satisfies the required precision of a 99% confidence interval with half-width not exceeding 1 minute.

The system operates continuously with ongoing arrivals and production, and is therefore modelled as a non-terminating system. Consequently, the primary performance measure is the steady-state mean cycle time of a laptop, defined as the total time a laptop spends in the

system from order acceptance to final completion, including all processing, waiting, and boxing stages. All time-based measures are expressed in minutes, and therefore the required precision corresponds to a half-width of 1 minute.

To obtain unbiased and statistically valid estimates, the simulation output is analyzed using the batch means method. Furthermore, the effect of initialization bias is addressed through the identification and removal of an appropriate warm-up period, ensuring that only steady-state observations are used in the analysis.

4.1 Treatment of Initialization Bias

Since the simulation model is initialized in an empty and idle state, the early observations are not representative of steady-state behavior. At the beginning of each replication, there are no queues and resources are immediately available, which leads to artificially low cycle times. Including these observations would therefore bias the results.

To address this, Welch's graphical procedure was applied. A total of 10 pilot replications were conducted, and the cycle times of completed laptops were recorded. The ensemble averages were computed across replications and smoothed using a moving average to reduce variability and reveal the underlying trend, as illustrated in Figure 20.

Figure 21 shows that the system initially experiences a start-up behaviour, where cycle times increase as congestion builds. After approximately 72 completed laptops, the curve stabilises and fluctuates around a constant level, indicating that steady-state has been reached. This corresponds to approximately 1240 minutes of simulation time.

Accordingly, a warm-up period of $T_0 = 1240$ minutes was selected, and all observations prior to this point were discarded. The remaining simulation run length was chosen to be sufficiently long relative to T_0 , ensuring that any residual initialisation bias is negligible.

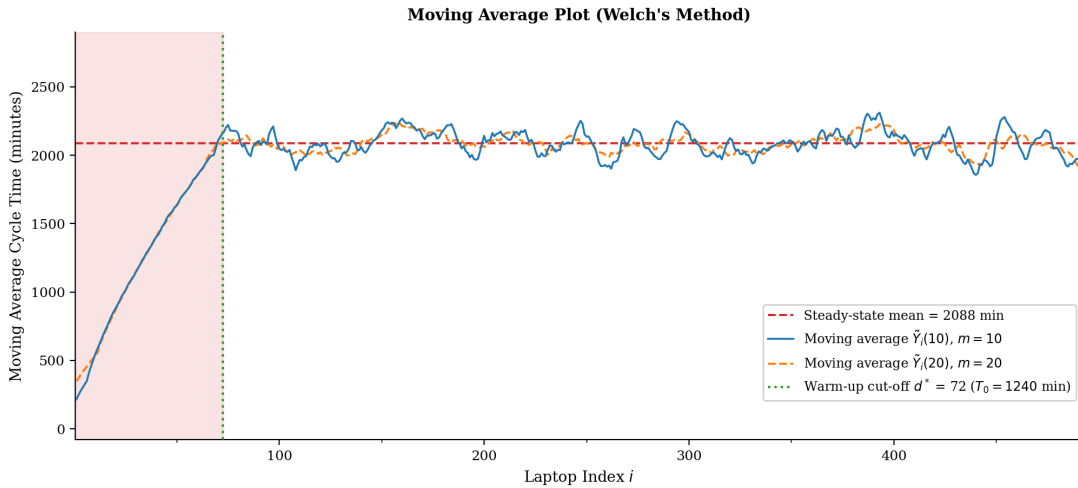


Figure 20: Welch's Moving Average Plot

4.2 Batch Means Estimation Framework

Following the removal of initialisation bias, the simulation output was analysed using the batch means method. A single long simulation run was performed, and the steady-state observations (after T_0) were divided into a number of non-overlapping batches of equal size. Initially, 20

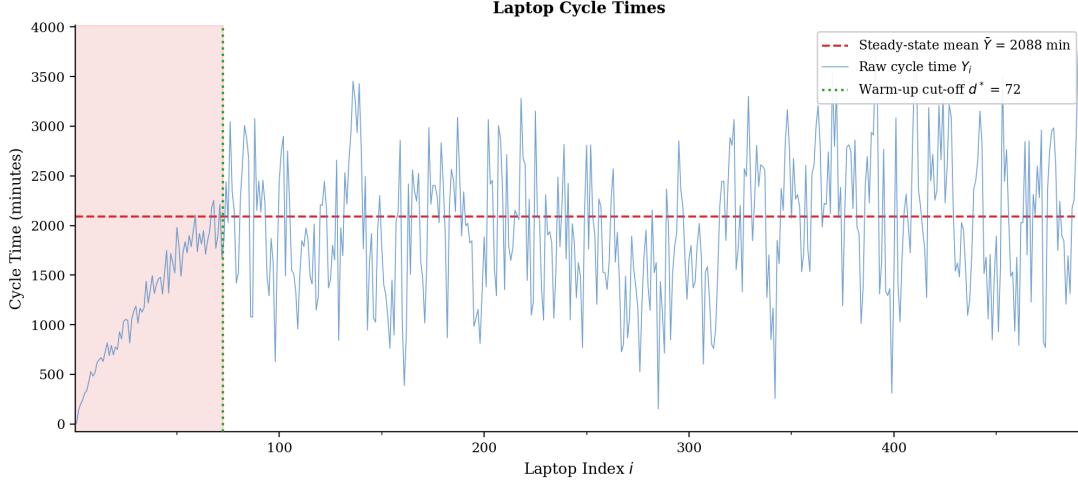


Figure 21: Laptop average cycle times from a single pilot run

batches were used to obtain pilot estimates of the mean and variance of the system. These batch averages were treated as approximately independent observations of the steady-state performance.

The batch means approach was adopted in preference to independent replications, as it avoids repeatedly discarding data during warm-up for each replication. Instead, the warm-up period is removed once, making more efficient use of simulation output while still allowing valid statistical inference based on the batch averages.

4.3 Confidence Interval and Precision Requirement

To estimate the steady-state mean cycle time and assess the required precision, a pilot study of $m_0 = 20$ batches was conducted. The average cycle time from each batch is summarised in Table 9. Based on these 20 pilot batches, the sample mean and standard deviation of the batch averages were $\bar{Y} = 2079.74$ minutes and $S = 54.49$ minutes. Using a 99% confidence interval, the strict requirement of a half-width not greater than 1 minute implies a very large number of batches, and is therefore computationally impractical. For this reason, a more practical relative precision criterion was adopted, with half-width limited to 1% of the estimated mean. This gives

$$\epsilon = 0.01 \times 2079.74 = 20.80,$$

and hence

$$m \geq \left(\frac{t_{m_0-1, 1-\alpha/2} \cdot S_0}{\epsilon} \right)^2 = \left(\frac{2.8609 \times 54.49}{20.80} \right)^2 \approx 56.25,$$

where $t_{m_0-1, 1-\alpha/2} = 2.8609$ is the critical value from the t -distribution with $m_0 - 1 = 19$ degrees of freedom at the 99% confidence level.

Therefore, 57 batches were used to obtain the final estimates of system performance. The resulting 99% confidence interval has a half-width of approximately 22.5 minutes, corresponding to a relative error of 1.083%, which is slightly above the 1% target.

Since the required number of batches is driven by the inherent variability of the system, achieving the exact target precision would require a disproportionate increase in computational effort. Therefore, 57 batches are considered sufficient, providing a near-target level of precision while maintaining practical feasibility. All subsequent results are based on these 57 batches.

Table 9: Average cycle times per batch

Rep	1	2	3	4	5	6	7	8	9	10
Cycle Time	1962.52	2132.80	2040.58	2111.91	2105.28	2041.36	2146.88	2073.82	2045.99	2135.39
Rep	11	12	13	14	15	16	17	18	19	20
Cycle Time	2065.05	2038.57	2143.48	2046.26	2044.55	2151.82	2023.78	2166.00	2086.41	2032.35

4.4 Baseline Results

The baseline simulation was run for 285,000 minutes (approximately 198 days) with 57 replications. The system completed 13,550 laptops from 34,935 arriving orders a fulfilment rate of only 38.8% with an average cycle time of 2,077.07 minutes (34.6 hours) and a standard deviation of 461.81 minutes. A maximum cycle time of 2,977.81 minutes confirms the system is highly variable. **End 1** recorded 30,823 rejected orders, meaning the 100 laptop capacity constraint was regularly reached, turning away 88.2% of all demand.

4.4.1 Resource Utilization

Type 1 workers are the most heavily utilized resource at 54.73%, serving six stages simultaneously. All other resources operate below 31%, suggesting performance limitations stem from raw material availability and synchronization delays rather than machine or worker capacity alone (Table 10).

Table 10: Baseline resource utilisation.

Resource	Capacity	Avg Use	Utilisation %
Type 1 Worker	7	3.831	54.73
Type 2 Worker	3	0.580	19.33
Type 3 Worker	3	0.577	19.24
Type 4 Worker	2	0.601	30.07
Type 5 Worker	2	0.284	14.20
MC_Roll Cut	2	0.380	19.02
MC_Special	3	0.577	19.24

4.4.2 Key Observations from Activities and Queues

The storage overflow activities (**Move to Storage** and **Transfer To Main Queue**) ran at 98.5% and 97.3% waiting respectively, confirming that Main Part Assembly overflow is almost permanently active. The inventory replenishment agents spent 14.5% (aluminium) and 39.1% (hard disks) of their time waiting to place orders, a clear sign of chronic stockouts. Keyboard Cutting processed 16,936 jobs for only 13,550 laptops, reflecting 3,386 rework passes (25% overhead). The two most critically congested queues are Aluminium Cutting (avg 86.13, max 100) and Wait Motherboard (avg 84.06, max 100) both stem from the same root cause described below.

4.4.3 KPI Summary

Table 11: Baseline KPI summary (285,000 min run, 57 replications).

Category	KPI	Value
Throughput	Orders arrived	34,935
	Orders rejected	30,823 (88.2%)
	Laptops completed	13,550
	Fulfilment rate	38.8%
	Average cycle time	2,077.07 min
	Std dev of cycle time	461.81 min
Critical Queues	Q Aluminium Cutting (avg / max)	86.13 / 100
	Q Wait Motherboard (avg / max)	84.06 / 100
	Q Quality Assurance (avg / max)	2.98 / 31
Resource Util.	Type 1 Worker	54.73%
	Type 4 Worker	30.07%
	All others	14-19%
Inventory	Aluminium replenishment jobs	270
	Hard disk replenishment jobs	193
Rework	Keyboard rework passes	3,386 (25% overhead)

4.4.4 Summary of Key Bottlenecks

The baseline results point to three interconnected bottlenecks:

1. **Aluminium and hard disk stockouts:** The Queue for Aluminium Cutting (avg 86.13) and the HardDisk Inventory minimum wait (217.14 min) confirm that both critical raw materials run out regularly, blocking production for extended periods.
2. **Downstream synchronization delay:** The Queue for waiting collection after finishing motherboard line (Queue for Wait Motherboard (avg 84.06)) is a direct consequence. When case components are delayed by blocked cutting due to stockout of aluminium blocks and reworks, finished motherboards wait for their matching partners at **Collect Parts**.
3. **Keyboard cutting rework:** Keyboard Cutting processed 16,936 jobs for only 13,550 laptops, generating 3,386 extra passes, a 25% overhead on a stage that shares MC.Special machine capacity and Type 3 workers with the screen shaping stage. This compounds the case line delay caused by aluminium stockouts, further widening the gap between case and motherboard completion times and increasing the wait at **Collect Parts**.
4. **Main Part Assembly storage overflow:** Move to Storage and Transfer To Main Queue ran at 98.5% and 97.3% waiting respectively, processing 2,101 jobs each incurring a 5-minute penalty (2 minutes to storage, 3 minutes to return). This means the Main Part Assembly queue capacity of 2 items is almost permanently exhausted, forcing a near-constant diversion through the storage path and adding avoidable delay to every affected motherboard item.

5 Model Improvement

Based on the simulation analysis, six improvement scenarios were identified and tested. For each recommendation, two or three parameter values were tested to identify diminishing returns and support a cost-benefit assessment. The same warm-up period and replication count as the baseline were used throughout to ensure comparability.

5.1 Reduce Raw Material Replenishment Lead Time

Under the current 900 minute lead time with a reorder point of $s = 10$, the factory regularly runs out of aluminium and hard disks before new stock arrives. This blocks cutting for extended periods and creates a cascade that simultaneously inflates the Queue for Aluminium Cutting (avg 86.13) and the Queue for Waiting Motherboard (avg 84.06). Reducing the lead time shortens each stockout event and requires no new equipment, only supplier negotiations or may be considering new suppliers who are close enough depending on cost tradeoff.

Table 12: Reduced lead time results summary.

Scenario	Lead Time (min)	Avg-Cycle Time (min)	% Change	Avg-Q Alum.	Avg-Q MB	Rejected Orders
Baseline	900	2,077.07	–	86.13	84.06	30,823
Scenario A	600	1,483.80	–28.6	84.09	82.36	29,271
Scenario B	450	1,186.49	–42.9	82.35	80.83	27,879
Scenario C	300	908.27	–56.3	79.64	78.43	25,762

Interpretation and Recommendation: Reducing the lead time produces substantial cycle time improvements across all three scenarios from 28.6% at 600 minutes up to **56.3%** at 300 minutes. Throughput also increases notably, rising from 13,550 laptops in the baseline to **18,950, 23,681, and 30,861** respectively. The reduction in rejected orders from **30,823 to 25,762** under Scenario C further demonstrates that faster replenishment keeps the system below the 100-laptop capacity cap more consistently, allowing more orders to be accepted.

However, the aluminium and motherboard queue sizes remain large even at 300 minutes (avg 79.6 and 78.4), confirming that stockouts still occur even with a halved lead time the reorder point of $s = 10$ remains too low. **This improvement is strongly recommended**, particularly Scenario C (300 minutes), which nearly halves the average cycle time and more than doubles throughput. It should ideally be combined with Recommendation from section 5.2 (raising the reorder point) to eliminate stockouts entirely.

5.2 Raise Raw Material Reorder Points (Q and s)

This improvement addresses the same root cause as Recommendation from section 5.1 (stockouts) but from a different angle: ordering earlier rather than waiting for faster delivery. Raising the reorder point creates a larger safety buffer during the lead time period. Unlike Recommendation from section 5.1, this requires no supplier negotiation and can be implemented immediately at virtually no cost. Three scenarios were tested, increasing both the reorder point and order quantity for aluminium and hard disks, as summarised in Table 13.

Table 13: Raised Raw Material Reorder points.

Scenario	Al- s	Al- Q	HDD- s	HDD- Q	Avg Cycle Time	% Change	Avg-Q Al.	Avg-Q MB
Baseline	10	50	10	70	2,077.07	–	86.13	84.06
Scenario A	20	60	20	90	1,730.79	-16.7%	83.17	85.12
Scenario B	30	70	30	100	1,482.34	-28.6%	84.78	82.86
Scenario C	40	80	40	120	1,296.07	-37.6%	87.09	87.44

Interpretation and Recommendation: Raising the reorder point to $s = 40$ (Scenario C) produced a 37.6% reduction in average cycle time and increased total throughput by approximately 60% compared to the baseline. While the queue sizes for aluminium cutting and waiting motherboards remain high due to the system handling much higher volume, the overall flow is significantly more efficient. This improvement is **strongly recommended** as the highest priority due to its zero implementation cost and immediate impact on the system’s primary bottlenecks. Improving the order quantity to a more reasonable amount (say 250 or 300) based on the storage availability will significantly have the better impact on the performance of the system.

5.3 Combined Inventory Policy

Recommendations from sections 5.1 and 5.2 each address the same root cause from different angles. Recommendation from section 5.1 shortens the duration of each stockout event by reducing the replenishment lead time, while Recommendation from section 5.2 prevents stockouts from occurring in the first place by ordering earlier. Combining both changes provides two-layer protection: the factory orders sooner and receives stock faster, virtually eliminating the extended production stoppages that drive the two largest queues in the system.

Table 14: Combined inventory policy parameters.

Scenario	Lead Time (min)	Al s	Al Q	HDD s	HDD Q
Baseline	900	10	50	10	70
Scenario A	450	30	70	30	100
Scenario B	300	40	80	40	120

Table 15: Simulation results.

Scenario	Avg-Cycle Time (min)	% Change	Avg-Q Alum.	Avg-Q MB	Avg-Q QA	Rejected Orders
Baseline	2,077.07	–	86.13	84.06	–	30,823
Scenario A	845.27	–59.3%	78.07	77.08	9.16	9,809
Scenario B	620.54	–70.1%	2.26	2.23	85.62	13,173

Interpretation and Recommendation: The combined policy produced substantial cycle time reductions across both scenarios. Scenario A (–59.3%) already outperforms Recommendation from section 5.1 in isolation (best: –56.3%), confirming that the two levers are complementary rather than substitutes. Scenario B achieves the strongest result in the entire study at –70.1%, cutting average cycle time from 2,077 min to just 621 min. The aluminium and motherboard

queues collapsed from 86 and 84 respectively to near zero in Scenario B (2.26 and 2.23), confirming that the root-cause stockout bottlenecks have been effectively eliminated.

Rejected orders fell sharply in Scenario A (from 30,823 to 9,809). The apparent rise in Scenario B (13,173) is an artifact of the simulation horizon, the faster throughput means more orders are still *in progress* at cut-off rather than timed-out, and the substantially higher number of completions (45,005 vs. 33,140) confirms the system is genuinely processing more work. The Quality Assurance (QA) queue rises sharply to **85.62** in Scenario B, indicating that once the inventory bottleneck is alleviated, the QA station emerges as the next binding constraint. This suggests that future capacity planning should prioritize QA expansion.

To address this emerging constraint, it is essential to expand Quality Assurance capacity by increasing the Type 4 staff from two to three workers. Adding a third worker is expected to reduce utilization to approximately 67%, effectively clearing the 541-minute backlog and ensuring the 70.1% front-end cycle time reduction successfully translates into higher completed order volume. In conclusion, **Scenario B of this combined policy is strongly recommended** as the highest-impact intervention available to management, provided that QA capacity is reviewed in parallel.

5.4 Reduce Keyboard Cutting Rework

This recommendation aims to reduce process waste by lowering the rework probabilities at keyboard cutting. Reducing these probabilities targets the 25% overhead in cutting jobs identified in the baseline.

Table 16: Recommendation 6: Scenario parameters and results.

Scenario	1st Prob.	2nd Prob.	Avg Cycle Time	% Change	Rework Count
Baseline	0.20	0.10	2,077.07	–	3,386
Scenario A	0.15	0.08	2,076.78	-0.01%	2,345
Scenario B	0.10	0.05	2,076.04	-0.05%	1,427
Scenario C	0.20	0	2,077.07	0.00%	3,386

Interpretation and Recommendation: Scenario B (probabilities 0.10/0.05) achieved the greatest reduction in waste, cutting rework events by 57.8% (from 3,386 to 1,427). Despite this substantial internal efficiency gain, the average cycle time only improved by 0.05%. This confirms that keyboard cutting is not the system’s primary constraint. These results reinforce the conclusion that process quality improvements should be a **secondary priority**, implemented only after addressing the severe inventory stockouts identified in Recommendation from section 5.2.

5.5 Additional Suggested Improvements

Beyond the above tested scenarios, the following design-level improvements are proposed as further opportunities to enhance system performance. While these were not tested in the simulation due to time constraints, each is grounded in the baseline results and supported by established operations management principles.

1. **Priority routing at the workbench:** Replacing FIFO with oldest-first routing at **Collect Parts** ensures accumulated dwell time is the primary driver for prioritization. This would reduce the maximum cycle time (currently 2,977 minutes) and compress the high standard deviation of 461 minutes, making delivery times more predictable for customers.

2. **Increase the WIP cap:** The baseline rejected 30,823 orders, 88% of all arrivals due to the 100-laptop capacity limit. Following the inventory improvements in Recommendations 2 and 3, laptops will move through the system faster and the WIP level will naturally stay lower. Raising the cap to 150 or 200 in this improved system would allow significantly more orders to be accepted without overwhelming production.
3. **Dedicated Type 1 workers per line:** Splitting the 7 Type 1 workers into dedicated pools for the motherboard and case lines would eliminate cross-line resource competition, reducing the variability in processing times that contributes to the high standard deviation of cycle time.
4. **Implement a pull (Kanban) system:** Triggering motherboard production only when a matching case component is ready would prevent the 84-item average queue at Q - Wait Motherboard, reduce WIP throughout the system, and improve synchronisation efficiency at Collect Parts.
5. **Increase queue capacities to eliminate storage overflow:** The Move to Storage activity ran at 98.5% waiting time in the baseline, processing 2,101 jobs each incurring a 5-minute penalty (2 minutes to storage, 3 minutes to return). Increasing the Power Assembly queue capacity from 3 to 5 and the Main Part Assembly queue from 2 to 4 would eliminate or significantly reduce this overflow in most cases, shortening the motherboard line path and reducing the average wait time at Q - Wait Motherboard. This is a zero-cost structural change that only requires modifying two queue capacity parameters in the model.
6. **Batch orders by size before production:** Currently orders of 1 laptop and orders of 6 laptops enter the system identically. Large orders take considerably longer to complete the boxing stage because all laptops in the order must pass Quality Assurance before boxing can begin, creating a blocking effect for smaller orders waiting behind them. Introducing a sorting gate that routes small orders (1–2 laptops) to a priority track ahead of large orders (4–6 laptops) would reduce the average cycle time for the majority of orders, improve customer experience for smaller purchasers, and reduce the high standard deviation of 461 minutes caused by large orders occupying the boxing stage for extended periods.

Conclusion

The baseline simulation revealed a severely constrained system, with 88.2% of orders rejected, an average cycle time of 2,077 minutes, and chronic stockouts driving two near-saturated queues at aluminium cutting and motherboard collection. Of the improvements tested, the combined inventory policy (Recommendation from section 5.3, Scenario B) delivered the greatest impact, reducing average cycle time by 70.1% to 621 minutes and collapsing the two critical queues to near zero. This came with the important caveat that Quality Assurance emerges as the next binding constraint under this scenario, with its queue rising to an average of 85.62, making QA capacity expansion an essential companion measure. Keyboard cutting rework reduction, while internally efficient, had a negligible effect on cycle time and should be treated as a secondary priority. Collectively, the results demonstrate that targeted inventory policy reform offers transformative gains at minimal cost, and that further improvements to queue capacities, worker allocation, and WIP limits represent viable next steps for continued system optimization.